KARAN DAVE, 260670122507, CSE B3, 2025-2029

ASSIGNMENT No: 15

TITLE: Write a Java program to demonstrate exception handling using finally, throw, and throws.

Problem Statement

Write a Java program to demonstrate exception handling using finally, throw, and throws.

Learning Objectives

To implement finally, throw, and throws for effective exception management.

Theory

The finally block always executes regardless of whether an exception occurs, making it suitable for resource cleanup. The throw keyword explicitly generates an exception, while throws declares exceptions that a method may pass to its caller. These mechanisms improve structured error handling and application stability.

Exercise

1. Create a Login program that throws an exception for invalid password and uses finally block.

```java
package assignment15;
import java.util.Scanner;
public class LoginDemo {
    public static void validateLogin(String username, String password) throws SecurityException {
        String validUser = "admin";
        String validPass = "SecurePass123";
        if (!username.equals(validUser) || !password.equals(validPass)) {
            throw new SecurityException("Authentication Failed: Invalid username or password.");
        } else {
            System.out.println("Login successful! Welcome, " + username + ".");
        }
    }
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.println("--- Login System ---");
        System.out.print("Enter Username: ");
        String user = scanner.nextLine();
        System.out.print("Enter Password: ");
        String pass = scanner.nextLine();
        try {
            validateLogin(user, pass);
        } catch (SecurityException e) {
            System.out.println("Error Caught: " + e.getMessage());
        } finally {
            System.out.println("[INFO] Login attempt process finalized. Closing connection resources.\n");
            scanner.close();
        }
    }
}
```

```
PS C:\Users\pdave\Downloads\Java sem 3\experiments\Assignment 15> java LoginDemo.java
--- Login System ---
Enter Username: admin
Enter Password: abc
Error Caught: Authentication Failed: Invalid username or password.
[INFO] Login attempt process finalized. Closing connection resources.

PS C:\Users\pdave\Downloads\Java sem 3\experiments\Assignment 15> java LoginDemo.java
--- Login System ---
Enter Username: admin
Enter Password: SecurePass123
Login successful! Welcome, admin.
[INFO] Login attempt process finalized. Closing connection resources.

PS C:\Users\pdave\Downloads\Java sem 3\experiments\Assignment 15>
```

2. Create an ATM PIN Verification program that throws an exception for an invalid PIN entered by the user and uses a finally block to display a message indicating that the verification process has completed.

```java
package assignment15;
import java.util.Scanner;
public class AtmPinDemo {
    private static final int CORRECT_PIN = 4321;
    public static void verifyPin(int enteredPin) throws IllegalArgumentException {
        if (enteredPin != CORRECT_PIN) {
            throw new IllegalArgumentException("Invalid PIN entered. Access Denied.");
        } else {
            System.out.println("PIN verified successfully. Please proceed with your transaction.");
        }
    }
    Run main | Debug main
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.println("--- ATM PIN Verification ---");
        System.out.print("Enter your 4-digit ATM PIN: ");
        int pin = 0;
        if (scanner.hasNextInt()) {
            pin = scanner.nextInt();
        } else {
            System.out.println("Invalid input format.");
            scanner.close();
            return;
        }
        try {
            verifyPin(pin);
        } catch (IllegalArgumentException e) {
            System.out.println("Exception Caught: " + e.getMessage());
        } finally {
            System.out.println("[INFO] The verification process has completed.");
            scanner.close();
        }
    }
}
```

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```
PS C:\Users\pdave\Downloads\Java sem 3\experiments\Assignment 15> javac AtmPinDemo.java
PS C:\Users\pdave\Downloads\Java sem 3\experiments\Assignment 15> java AtmPinDemo.java
--- ATM PIN Verification ---
Enter your 4-digit ATM PIN: 1111
Exception Caught: Invalid PIN entered. Access Denied.
[INFO] The verification process has completed.
PS C:\Users\pdave\Downloads\Java sem 3\experiments\Assignment 15> java AtmPinDemo.java
--- ATM PIN Verification ---
Enter your 4-digit ATM PIN: 4321
PIN verified successfully. Please proceed with your transaction.
[INFO] The verification process has completed.
PS C:\Users\pdave\Downloads\Java sem 3\experiments\Assignment 15>
```